Proposed Solution Report

SMARTBRIDGE - Project Proposal (Proposed Solution) Report

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Title:** AI FitTrack - AI-Powered Personalized Fitness & Workout Recommendation System
- **Maximum Marks:** 5 Marks

Executive Summary

The AI FitTrack proposal report aims to transform workout tracking and fitness guidance using Node.js, Express, MongoDB, and Google Gemini AI. It tackles manual logging inaccuracies and lack of personalized training guidance, promising structured operations, secure JWT data privacy, and actionable AI workout insights. Key features include RESTful workout management, full-text history search, and real-time Gemini AI recommendation processing.

Project Overview

- **Objective:** The primary objective is to revolutionize manual exercise tracking by deploying a secure, scalable MVC REST backend integrated with Google Gemini AI (gemini-2.5-flash) for real-time, personalized workout recommendations and performance analytics.
- **Scope:** The project comprehensively provides complete CRUD workout management, secure user registration/authentication, search filtering, and AI-driven workout recommendations and insights.

Problem Statement

- **Description:** Fitness enthusiasts struggle with manual record-keeping, inaccurate exercise histories, lack of data security, and an inability to get personalized fitness advice without expensive trainers.
- **Impact:** Resolving these issues provides centralized workout storage, saves tracking time, enhances user data security via JWT encryption, and delivers intelligent AI training plans.

Proposed Solution

- **Approach:** Implementing a Node.js/Express RESTful backend coupled with MongoDB Atlas and Google Gemini AI service layers to evaluate user metrics (age, goals, experience level) and generate customized fitness plans.
- **Key Features:**
    - JWT-based secure user authentication and bcrypt password encryption.
    - Full RESTful CRUD operations for workouts and search filtering.
    - Google Gemini AI integration (@google/genai) for personalized workout generation and fitness insights.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel Core i5 / AMD Ryzen 5 (8th Gen or above)
	Memory (RAM)	8 GB minimum (16 GB recommended)
	Storage	1 GB available disk workspace
Software	Frameworks & Runtime	Node.js (v16+), Express.js
	Libraries	Mongoose ODM, bcryptjs, jsonwebtoken, @google/genai, dotenv, cors
	Development Environment	Visual Studio Code, Postman / Thunder Client
Data	Data Source / Collections	MongoDB Atlas Cloud Database (Users & Workouts collections)